Technical Specifications: HackRF One

Radiofrequency Engineering

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Contents

1 Gains

Controlled separately through `libhackrf` (tools such as `hackrf_transfer` or GNU Radio):

- **LNA (IF RX):** `hackrf_set_lna_gain()` - 0 to 40 dB, adjustable in 8 dB steps. Affects the signal in the intermediate-frequency (IF) stage.
- **VGA (Baseband RX):** `hackrf_set_vga_gain()` - 0 to 62 dB, adjustable in 2 dB steps. Variable-gain amplifier in baseband.
- **TX VGA:** `hackrf_set_txvga_gain()` - 0 to 47 dB, adjustable in 1 dB steps.
- **AMP (RF Front End):** `hackrf_set_amp_enable()` - 0 (off) or 1 (on). Enables the RF front-end amplifier, providing a fixed ≈+14 dB.

  > The amplifier part number varies depending on the hardware revision. The MGA-81563 appears in third-party schematics for early revisions, but it is not confirmed as an official specification in the Great Scott Gadgets documentation.

  Applies to the final TX stage or the initial RX stage.

Note: Since the ADC is 8-bit (dynamic range limited to ≈48 dB), correct gain management is crucial to avoid raising the noise floor or causing clipping during analog-to-digital conversion.

2 Hardware Filters and Processing

- **Analog Filter (Baseband LPF):** `hackrf_set_baseband_filter_bandwidth()` Selectable thresholds in the MAX2837 (r1–r8) / MAX2839 (r9) transceiver: 1.75, 2.5, 3.5, 5, 5.5, 14, 20, 24, and 28 MHz. Helps prevent analog aliasing.
- **CPLD and Microcontroller (No Internal DSP):** Uses a Xilinx XC2C64A CPLD for signal routing and an ARM Cortex-M4 MCU (NXP LPC4320/30). Unlike other high-end SDRs, the HackRF One **does not perform internal digital signal processing** (such as DDC/DUC or hardware decimation). All mathematical processing falls to the host CPU over USB.

3 Frequency Ranges

- **Nominal Range:** 1 MHz to 6 GHz.
- **Synthesis Architecture:**
  - 2.3 GHz to 2.7 GHz: Direct conversion using the MAX2837/MAX2839 transceiver.
  - Outside that band: Uses the RFFC5071 mixer to perform up-conversion or down-conversion into the transceiver band.
- **Performance note:** Below 10 MHz, the efficiency of the RF transformers (baluns) decreases.

4 Sample Rate and Resolution

Limited by the ADC/DAC (MAX5864) capability and the USB 2.0 bus.

Note: To avoid packet drops (dropped samples) at 20 MS/s, it is recommended to connect the HackRF to a dedicated USB controller on the host that does not share bandwidth with other high-demand peripherals. Rates below 2 MS/s are not officially supported.

Mode	I/Q Resolution	Sample Rate	Maximum Bandwidth
Standard	8-bit I / 8-bit Q	2 to 20 MS/s	20 MHz (theoretical Nyquist)

5 IQ & DC Corrections

- **DC Peak (DC Offset / LO Leakage):** Due to its homodyne architecture (Zero-IF / Direct Conversion), the HackRF One shows a visible spike at the center of the spectrum (0 Hz in baseband). It **must be corrected in software** (e.g., *DC Blocker* in GNU Radio) or by using *Offset Tuning* (tuning a few MHz to the side and digitally shifting).
- **I/Q Imbalance:**

  It can appear as ghost mirror frequencies when working with strong signals. It is recommended to calibrate IQ gain and phase in the host software (SDR#, GNU Radio), since the device does not apply automatic factory corrections.

6 Connectors and Inputs

Connector	Function	Specification
Micro-USB	Power and data	USB 2.0 High Speed (480 Mbps). 5 V / ≈500 mA.
ANT (SMA)	Main TX/RX	Female, 50 Ω. Half-duplex. 1 MHz – 6 GHz.
CLKIN (SMA)	External reference	10 MHz. Square wave, **0 V to 3.0 V**. Do not exceed 3.0 V or go below 0 V. Do not use other frequencies except with firmware modification.
CLKOUT (SMA)	Output clock	10 MHz, 3.3 V. Enabled with `hackrf_clock -o 1`. For daisy chaining between units.
P20, P22, P28	Expansion headers	LPC4320 MCU pins. 3.3 V logic, for modules such as PortaPack.

7 Safe Operating Limits

Critical warnings - LNA / front-end susceptibility

- **Maximum RX Power:** NEVER exceed **-5 dBm** (≈0.3 mW) at the antenna port. Exceeding it will destroy the front-end amplifier. In theory, with the AMP disabled it may tolerate up to +10 dBm, but a software error that re-enables it would cause permanent damage - always use an external attenuator.
- **TX power by band** (reference for sizing loopback paths and amplifiers):
  - 1–10 MHz: 5 to 15 dBm (increases with frequency).
  - 10–2170 MHz: 5 to 15 dBm (decreases with frequency).
  - 2170–2740 MHz: 13 to 15 dBm (best performance region).
  - 2740–4000 MHz: 0 to 5 dBm.
  - 4000–6000 MHz: -10 to 0 dBm.

In TX→RX loopback: use sufficient attenuation for the band. Never connect TX directly to RX.

- **Unloaded RF port:** Do not transmit without an antenna or a 50 Ω dummy load. Reflected power can burn out the TX stage.
- **Clock Voltages:** Signals on CLKIN must not be negative or exceed **3.0 V**. A sine wave centered at 0 V requires DC offset to meet this requirement.
- **Antenna Power (Bias-T):** Can provide 3.3 V at 50 mA on the SMA connector to power external LNAs. Do not enable it if the connected device presents a DC short circuit.

8 Clock and Synchronization

- **Internal Oscillator:** Standard 10 MHz crystal. Ppm stability is not officially specified by Great Scott Gadgets; community estimates suggest 20–30 ppm, susceptible to thermal drift.
- **TCXO Module (External/Additional):** For serious use at high frequencies or with narrow modulation, attach a 0.5 ppm TCXO module on header P22 or use CLKIN with a stable external reference.
- **CLKIN:**
  - 10 MHz signal, 0 to 3.0 V square wave. High impedance.
  - The device automatically switches to the external clock when the next TX/RX operation starts. Verify detection with `hackrf_debug --si5351c -n 0 -r` (output `0x01` = clock detected).
  - CLKOUT from one HackRF One can be directly daisy chained to CLKIN on another.
- **CLKOUT:** Enable with `hackrf_clock -o 1`. Allows synchronization of multi-HackRF One arrays with a single master clock.

9 Indicator LEDs

Six LEDs are mounted along the board. When connected to USB, **four should turn on**: 3V3, 1V8, RF, and USB.

Note: Adjacent LED colors are different to make visual distinction easier, but the colors themselves have no functional meaning.

10 Expansion Pinout (P20 / P22 / P28)

Provides access to microcontroller GPIOs (designed to interconnect add-ons such as the PortaPack). They operate at **3.3 V**.

P20 and P28

- Contain pins for the LPC internal ADC, I2C, SPI, and auxiliary power.
- They are the backbone of parallel communication for sending IQ samples to a touchscreen (PortaPack) instead of going over USB.

LED	Function	State and interpretation
3V3	Main power	On when USB is connected. Indicates that the 3.3 V regulator is working. If it does not turn on: hardware fault.
1V8	Active firmware + secondary power	On when USB is connected. Indicates that the firmware is running and has enabled additional internal power rails (1.8 V). If it does not turn on while 3V3 is on: firmware problem.
RF	RF chain powered	On when USB is connected. Indicates that the firmware has enabled power to the RF block.
USB	Active USB communication	On when the host recognizes the device correctly.
RX	Reception in progress	On **only** during active captures.
TX	Transmission in progress	On **only** during active transmission.

P22

- Clock header and bus expansion signals.
- This is where the **0.5 ppm TCXO** modules are installed. When inserted, the primary clock changes to the one from this header, stabilizing the general PLL.
- Also exposes alternate CLKIN (pin 2), enabled with `hackrf_clock -c p22`.

Usage note: Do not exceed the LPC43xx current thresholds on any pin. Exceeding them will permanently damage the central digital logic.